

final_CPU

Vivado Synthesized Design - Utilization Report

Tool Version	Vivado v.2024.2 (win64) Build 5239630
Design	final_CPU
Device	xc7k70tfbv676-1
Design State	Synthesized
Date	Mon Sep 7 00:24:19 2026

Command: report_utilization -file final_CPU_utilization_synth.rpt -pb final_CPU_utilization_synth.pb

1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	0	0	0	41000	0.00%
LUT as Logic	0	0	0	41000	0.00%
LUT as Memory	0	0	0	13400	0.00%
Slice Registers	0	0	0	82000	0.00%
Register as Flip Flop	0	0	0	82000	0.00%
Register as Latch	0	0	0	82000	0.00%
F7 Muxes	0	0	0	20500	0.00%
F8 Muxes	0	0	0	10250	0.00%

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	-	-
0	—	-	Set
0	—	-	Reset
0	—	Set	-
0	—	Reset	-
0	Yes	-	-
0	Yes	-	Set
0	Yes	-	Reset
0	Yes	Set	-

Total	Clock Enable	Synchronous	Asynchronous
0	Yes	Reset	-

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	135	0.00%
RAMB36/FIFO*	0	0	0	135	0.00%
RAMB18	0	0	0	270	0.00%

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1.

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	240	0.00%

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	0	0	0	300	0.00%
Bonded IPADs	0	0	0	26	0.00%
Bonded OPADs	0	0	0	16	0.00%
PHY_CONTROL	0	0	0	6	0.00%
PHASER_REF	0	0	0	6	0.00%
OUT_FIFO	0	0	0	24	0.00%
IN_FIFO	0	0	0	24	0.00%
IDELAYCTRL	0	0	0	6	0.00%
IBUFDS	0	0	0	288	0.00%
GTXE2_COMMON	0	0	0	2	0.00%
GTXE2_CHANNEL	0	0	0	8	0.00%
PHASER_OUT/PHASER_OUT_PHY	0	0	0	24	0.00%
PHASER_IN/PHASER_IN_PHY	0	0	0	24	0.00%
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	300	0.00%
ODELAYE2/ODELAYE2_FINEDELAY	0	0	0	100	0.00%
IBUFDS_GTE2	0	0	0	4	0.00%
ILOGIC	0	0	0	300	0.00%
OLOGIC	0	0	0	300	0.00%

5. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	0	0	0	32	0.00%
BUFIO	0	0	0	24	0.00%

Site Type	Used	Fixed	Prohibited	Available	Util%
MMCME2_ADV	0	0	0	6	0.00%
PLLE2_ADV	0	0	0	6	0.00%
BUFMRCE	0	0	0	12	0.00%
BUFHCE	0	0	0	96	0.00%
BUFR	0	0	0	24	0.00%

6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00%
CAPTUREE2	0	0	0	1	0.00%
DNA_PORT	0	0	0	1	0.00%
EFUSE_USR	0	0	0	1	0.00%
FRAME_ECCE2	0	0	0	1	0.00%
ICAPE2	0	0	0	2	0.00%
PCIE_2_1	0	0	0	1	0.00%
STARTUPE2	0	0	0	1	0.00%
XADC	0	0	0	1	0.00%

7. Primitives

Ref Name	Used	Functional Category
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8. Black Boxes

Ref Name	Used
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9. Instantiated Netlists

Ref Name	Used
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Source report reproduced from the supplied Vivado utilization output. No values have been changed.